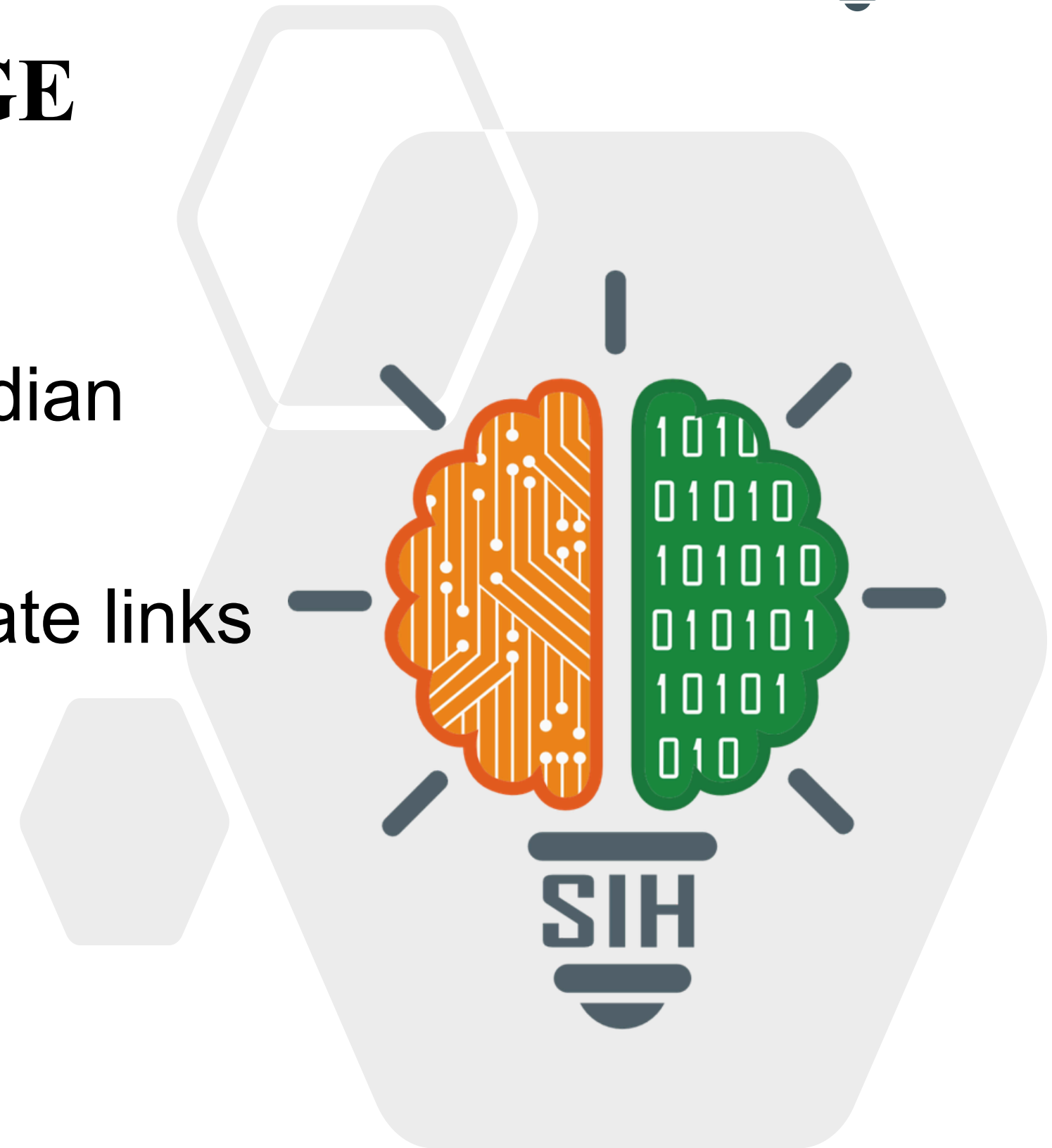


TITLE PAGE

- **Problem Statement ID** – SIH26173
- **Problem Statement Title**- iTantra -Indian Multilingual TTS & STT Aided Neural Transceiver Radio Access for low bitrate links
- **Theme**- Miscellaneous
- **PS Category**- Software
- **Team ID**- Team s46
- **Team Name** - KABOOTAR



iTantra — an offline, multilingual neural voice transceiver designed for communication over low-bandwidth links.



DETAILED EXPLANATION



Tap to Speak

iTantra is an offline, multilingual voice transceiver designed for low-bandwidth environments that converts speech into lightweight text for transmission and reconstructs it as voice at the receiver, enabling fast, reliable communication without internet or cloud services.



INNOVATION AND UNIQUENESS



Speech-to-Speech Pipeline: Converts Speech → Text → Transmission → Speech to reduce communication data.



Offline AI: STT and TTS run locally on the device, eliminating cloud/API dependency.



Low-Bandwidth Communication: Transmits text instead of raw audio, making voice communication feasible over constrained links.



Device-to-Device: Enables direct communication using Wi-Fi/Bluetooth/Radio signals even without internet.



Emergency Ready: Walkie-Talkie like features + emergency alert functionality



ADDRESSING THE PROBLEM

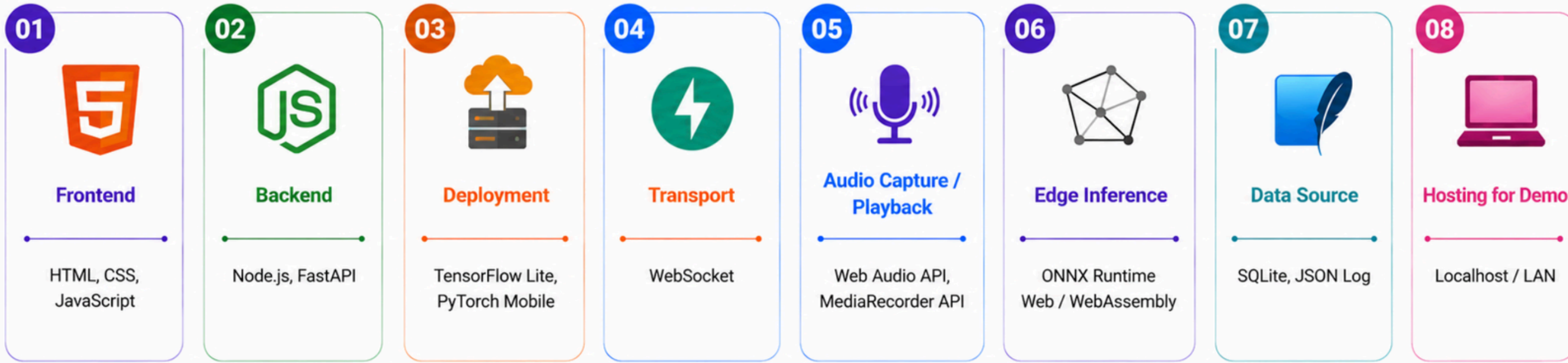
Problem: Vocal audio is highly data-intensive, making communication unreliable over low-data-rate links, especially during disasters, remote operations and network-constrained situations.

iTantra solves this by:

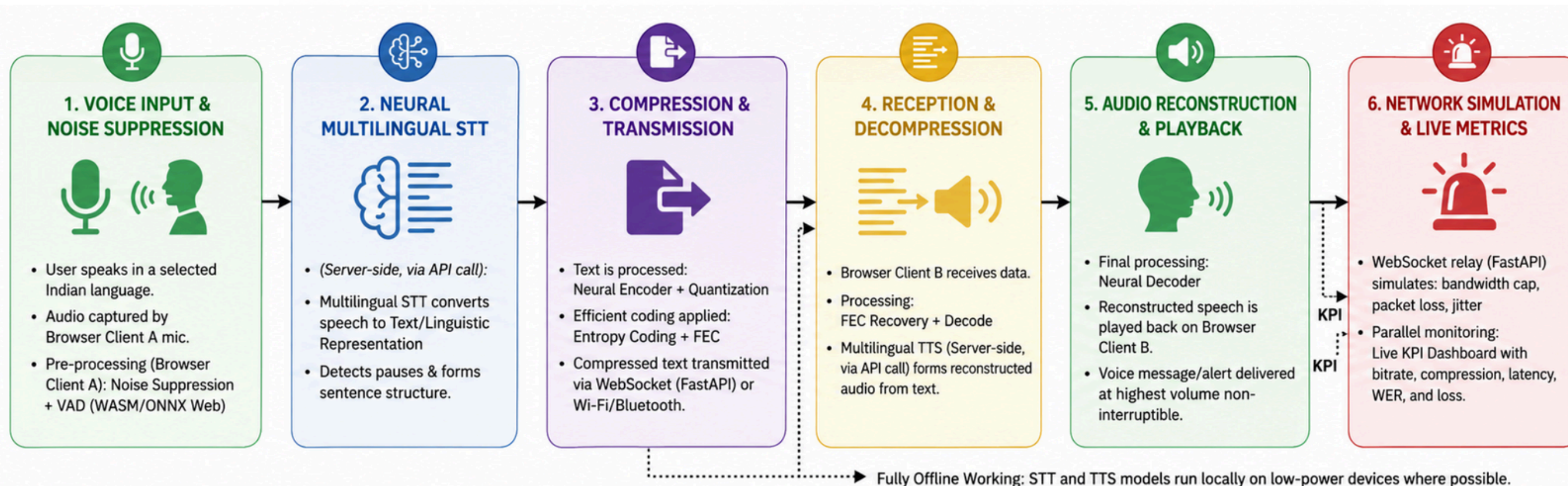
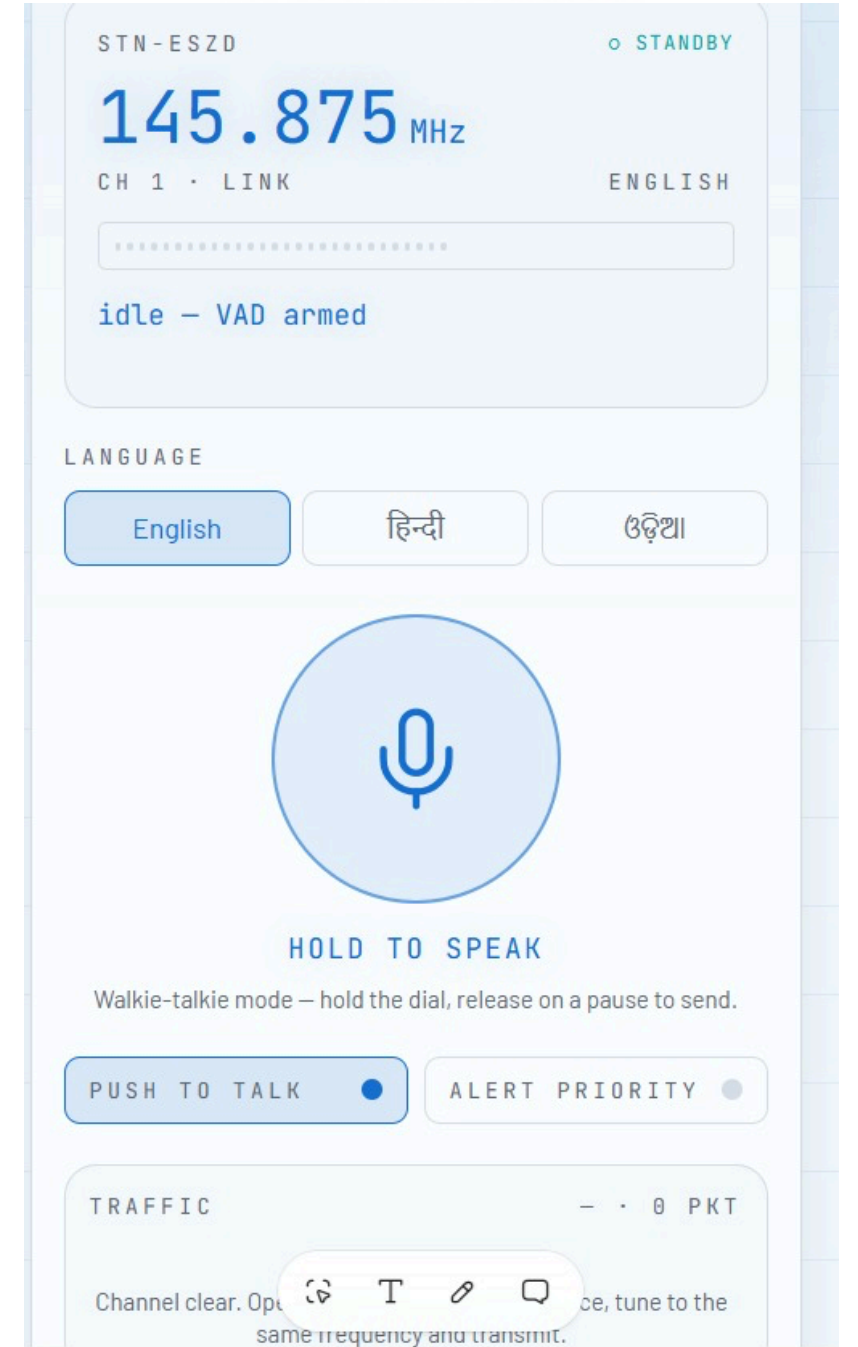


This dramatically reduces bandwidth requirements.

• TECH STACK •



• METHODOLOGY •

CURRENT
PROTOTYPE



FEASIBILITY



1. Technical Feasibility:

- Lightweight on-device AI models enable STT/TTS on low and mid-range smartphones.
- Uses existing Wi-Fi, Bluetooth and Radio hardware without specialized equipment.



2. Operational Feasibility:

- Simple Speech → Text → Transfer → Speech workflow enables real-time communication.
- Supports walkie-talkie like features and emergency communication scenarios.



3. Economic Feasibility:

- Low deployment cost by utilizing existing smartphone hardware.
- Reduced dependency on mobile data and cloud infrastructure.

POTENTIAL CHALLENGES
AND RISKS

1. Device Limitations:

- Limited processing power, memory and battery on low-end devices.



2. Speech Recognition Accuracy:

- Accents, background noise and regional language variations may affect STT accuracy.



3. Connectivity & Range:

- Wi-Fi/Bluetooth/Radio communication may have limited range and intermittent connectivity.

STRATEGIES FOR OVERCOMING
THESE CHALLENGES

1. AI Optimization:

- Use quantization and compressed models to reduce memory, latency and power consumption.



2. Improved Recognition:

- Apply noise filtering and language-specific models to improve speech accuracy.



3. Reliable Communication:

- Implement automatic reconnection and adaptive switching between Wi-Fi, Bluetooth and Radio Signal.



4. Scalable Design:

- Use a modular architecture to easily add languages, devices and communication modes.

IMPACT

**RURAL & LOW-CONNECTIVITY USERS**

Communication with limited/no internet.

**REGIONAL LANGUAGE USERS**

Breaks language barriers with different Indian languages.

**EMERGENCY & DISASTER RESPONSE**

Communication during network failures and emergencies.

**LOW-END DEVICE USERS**

Affordable communication without specialized hardware.

**STUDENTS & FIELD TEAMS**

Enables walkie-talkie style features for communication in campuses, events and remote areas.

BENEFITS

**SOCIAL BENEFIT**

- Promotes digital and linguistic inclusion
- Improves communication accessibility in remote communities
- Supports communication during emergencies

**ENVIRONMENTAL BENEFIT**

- Reduces the need for dedicated communication hardware
- Optimizes device resources through lightweight processing
- Enables reuse of existing smartphones instead of specialized devices

**ECONOMIC BENEFIT**

- Reduces dependence on costly telecom infrastructure
- Works on low/mid-range devices, lowering adoption cost
- Minimizes mobile-data usage through lightweight text transmission
- Can be deployed for institutions, disaster-response teams and field operations

**DISASTER & PUBLIC SAFETY**

- Provides an alternative communication channel during network outages.
- Helps rescue workers, volunteers and relief teams to quickly exchange critical information and instructions.
- Supports rapid, short-range communication without internet dependency.

TARGET AUDIENCE



**ILLITERATE &
NON-LITERATE USERS**



**LOW-BANDWIDTH
NETWORK USERS**



**MULTILINGUAL
COMMUNITIES**



**DISASTER & EMERGENCY
RESPONDERS**

Speech-to-Text (STT) — Offline / On-device:

- **AI4Bharat / Vakyansh ASR Toolkit (IIT Madras)** — 40+ Indic language ASR models, NeMo-based, Apache-licensed — github.com/Open-Speech-EkStep/vakyansh-models
- **whisper.cpp** — quantized on-device inference of OpenAI Whisper, works fully offline on mobile CPUs — github.com/ggerganov/whisper.cpp

Text-to-Speech (TTS) — Offline / On-device:

- **AI4Bharat Indic-TTS** — open-source TTS (FastPitch + HiFi-GAN) for 13 Indic languages — github.com/AI4Bharat/Indic-TTS
- **sherpa-onnx (k2-fsa)** — runs VITS-family TTS models on-device on Android/embedded hardware, low CPU footprint — github.com/k2-fsa/sherpa-onnx

Datasets:

- **AI4Bharat IndicCorp / IndicVoices** — large-scale multilingual speech + text corpus



- **IITM Indic-TTS Dataset** — per-language studio-quality speech corpus used for TTS training